

§7.4 - Integration of Rational Functions by Partial Fractions

Definition #1 (Polynomial)

Let $a_0, a_1, \dots, a_n \in \mathbf{R}$ where $a_n \neq 0$.

Then

$$P(x) = a_0 + a_1x + \dots + a_nx^n$$

is a **polynomial** of degree n .

If $n = 1$ then P is a linear polynomial.

We have

$$P(x) = 5x - 3$$

linear, and

$$Q(x) = x^3 - 1$$

is of degree 3.

Definition #2 (Reducible Polynomial)

Let P be a polynomial of degree n .

If u and v are polynomials of degree less than n such that

$$P(x) = u(x) \cdot v(x)$$

then P is a **reducible polynomial**. Otherwise, P is an **irreducible polynomial**.

We have

$$Q(x) = x^3 - 1 = (x - 1)(x^2 + x + 1),$$

so it is reducible. While

$$P(x) = x^2 + 1$$

is irreducible, it has no real roots.

If P is of degree 2 or more and $P(x) = 0$ for some $x \in \mathbb{R}$, then P is reducible.

Definition #3 (Roots)

Let P be a polynomial and $x_0 \in \mathbf{R}$.

If $P(x_0) = 0$ then there exists a $k \in \mathbf{N}$ and a polynomial Q such that

$$P(x) = (x - x_0)^k Q(x), \quad Q(x_0) \neq 0.$$

We say x_0 is a **root** of P with **order** k .

If $k = 1$ then x_0 is a **simple root** and if $k > 1$ we have a **repeated root**.

For

$$P(x) = (x^2 + 1)(x - 5)(x - 3)^7$$

we have a simple root at 5 and a 7th order root at 3.

Definition #4 (Rational Function)

Let P and Q be polynomials.

Then R given by

$$R(x) = \frac{P(x)}{Q(x)}$$

is a **rational** function.

If the degree of Q is greater than the degree of P then R is **proper rational function**.

Any proper rational function can be written as a sum of partial fraction terms.

Proper rational:

$$\frac{-x + 2}{x^2 + 3x + 2}$$

Improper rational:

$$\frac{2x^2 + 5x + 6}{x^2 + 3x + 2}$$

Definition #5 (Partial Fraction Decomposition)

Let R be a proper rational function.

Then the **partial fraction decomposition** of R is

$$R(x) = \sum_{k=1}^m \frac{P_k(x)}{Q_k(x)}$$

where P_k is a polynomial and Q_k is a power of an irreducible polynomial.

(Here the degree of Q_k is greater than the degree of P_k for all k .)

Partial fractions:

$$\frac{3}{x+2} \quad \text{and} \quad \frac{3x+1}{x^2+1}$$

Not a partial fraction:

$$\frac{-x+2}{x^2+3x+2} = \frac{-x+2}{(x+2)(x+1)}$$

Example #1 (Simple roots)

Find the partial fraction decomposition of

$$\frac{x - 3}{(x - 2)(x + 4)}.$$

Guess the form:

$$\begin{aligned}\frac{x-3}{(x-2)(x+4)} &= \frac{A}{x-2} + \frac{B}{x+4} \\ &= \frac{A(x+4)}{(x-2)(x+4)} + \frac{B(x-2)}{(x-2)(x+4)}\end{aligned}$$

so

$$\Rightarrow x-3 = A(x+4) + B(x-2).$$

If $x = 2$

$$\begin{aligned}2-3 &= A \cdot 6 + B \cdot 0 \\ \Rightarrow -1 &= 6A \\ \Rightarrow A &= -\frac{1}{6}.\end{aligned}$$

If $x = -4$

$$\begin{aligned}-4-3 &= A \cdot 0 + B(-4-2) \\ \Rightarrow -7 &= -6B \\ \Rightarrow B &= \frac{7}{6}.\end{aligned}$$

Then

$$\frac{x-3}{(x-2)(x+4)} = -\frac{1}{6} \cdot \frac{1}{x-2} + \frac{7}{6} \cdot \frac{1}{x+4}$$

is easy to integrate.

Example #2

Find

$$\int \frac{x - 3}{(x - 2)(x + 4)} dx.$$

We have

$$\int \frac{x-3}{(x-2)(x+4)} dx = -\frac{1}{6} \int \frac{1}{x-2} dx + \frac{7}{6} \int \frac{1}{x+4} dx$$
$$= -\frac{1}{6} \ln|x-2| + \frac{7}{6} \ln|x+4| + C.$$

Example #3 (Repeated roots)

Find the partial fraction decomposition of

$$\frac{x - 4}{(x - 2)(x + 4)^2}.$$

Guess the form:

$$\frac{x-4}{(x-2)(x+4)^2} = \frac{A}{x-2} + \frac{B}{x+4} + \frac{C}{(x+4)^2}$$

so

$$x-4 = A(x+4)^2 + B(x-2)(x+4) + C(x-2).$$

If $x = 2$

$$-2 = 36A + 0 + 0$$

$$\Rightarrow A = -\frac{1}{18}.$$

If $x = -4$

$$-8 = 0 + 0 - 6C$$

$$\Rightarrow C = \frac{4}{3}$$

We need B , so choose $x = 4$ (or any x other than -4 or 2)

$$0 = 64A + 16B + 2C$$
$$\Rightarrow 16B = -64A - 2C$$

$$B = \frac{1}{16} \left(64 \cdot \frac{1}{18} - \frac{8}{3} \right)$$
$$= \frac{1}{16} \left(\frac{32}{9} - \frac{24}{9} \right)$$
$$= \frac{1}{16} \cdot \frac{8}{9}$$
$$= \frac{1}{18}$$

Hence

$$\frac{x-4}{(x-2)(x+4)^2} = -\frac{1}{18} \cdot \frac{1}{x-2} + \frac{1}{18} \cdot \frac{1}{x+4} + \frac{4}{3} \cdot \frac{1}{(x+4)^2}$$

is easy to integrate.

Example #4

Find

$$\int \frac{x - 4}{(x - 2)(x + 4)^2} dx.$$

Here

$$\begin{aligned}\int \frac{x-4}{(x-2)(x+4)^2} dx &= -\frac{1}{18} \int \frac{1}{x-2} dx + \frac{1}{18} \int \frac{1}{x+4} dx \\ &\quad + \frac{4}{3} \int \frac{1}{(x+4)^2} dx \\ &= -\frac{1}{18} \ln|x-2| + \frac{1}{18} \ln|x+4| \\ &\quad - \frac{4}{3} \frac{1}{x+4} + C\end{aligned}$$

since

$$\int \frac{1}{(x+4)^2} dx = \int \frac{1}{u^2} du = -\frac{1}{u} + C = -\frac{1}{x+4} + C.$$

Example #5 (Simple irreducible factor)

Find the partial fraction decomposition of

$$\frac{3x - 1}{(x - 1)(x^2 + 2x + 5)}.$$

Guess the form:

$$\frac{3x-1}{(x-1)(x^2+2x+5)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2x+5}$$

$$\begin{aligned} 3x-1 &= A(x^2+2x+5) + (Bx+C)(x-1) \\ &= A(x^2+2x+5) + B(x^2-x) + C(x-1) \end{aligned}$$

If $x = 1$

$$\begin{aligned} 2 &= 8A \\ \Rightarrow A &= \frac{1}{4} \end{aligned}$$

In general,

$$0x^2 + 3x - 1 = A(x^2 + 2x + 5) + B(x^2 - x) + C(x - 1).$$

Equate 2nd order terms

$$0 = A + B \Rightarrow B = -\frac{1}{4}$$

Equate 0th order terms

$$\begin{aligned} -1 &= 5A - C \\ \Rightarrow C &= \frac{9}{4} \end{aligned}$$

(We could have also used our technique from earlier and picked $x = 0$ or -1 to solve for B & C .)

We find

$$\frac{3x-1}{(x-1)(x^2+2x+5)} = \frac{1}{4} \cdot \frac{1}{x-1} + \frac{1}{4} \cdot \underbrace{\frac{-x+9}{x^2+2x+5}}_{\text{easy to integrate?}}$$

Example #6

Find

$$\int \frac{-x + 9}{x^2 + 2x + 5} dx$$

With

$$u = x^2 + 2x + 5 \quad du = (2x + 2)dx$$

we have the wrong constant term in our numerator for u -sub:

$$A(2x + 2) \neq -x + 9 \text{ for all } A \in \mathbb{R}.$$

So toss the bad terms out! Want

$$-x + 9 = A(2x + 2) + B$$

for a "clean" u -sub. Take

$$-x + 9 = -\frac{1}{2}(2x + 2) + 10$$

(Finding A and B is just like solving for the earlier systems for the PFDs)

Then

$$\frac{-x + 9}{x^2 + 2x + 5} = -\frac{1}{2} \cdot \underbrace{\frac{2x + 2}{x^2 + 2x + 5}}_{\substack{\text{easy} \\ u\text{-sub}}} + \frac{10}{x^2 + 2x + 5}.$$

How to compute

$$\int \frac{10}{x^2 + 2x + 5} dx?$$

First, complete the square,

$$(x + 1)^2 = x^2 + 2x + 1$$

$$\Rightarrow (x + 1)^2 + 4 = x^2 + 2x + 5$$

and factor some more

$$\begin{aligned}\int \frac{10}{x^2 + 2x + 5} dx &= 10 \int \frac{1}{(x+1)^2 + 4} dx \\ &= \frac{10}{4} \int \frac{1}{\left(\frac{x+1}{2}\right)^2 + 1} dx.\end{aligned}$$

Now choose

$$w = \frac{x+1}{2} \quad dw = \frac{1}{2} dx \Rightarrow dx = 2dw$$

$$\begin{aligned}\int \frac{10}{x^2 + 2x + 5} dx &= \frac{10}{4} \cdot 2 \int \frac{1}{w^2 + 1} dw \\ &= 5 \arctan(w) + C_1 \\ &= 5 \arctan\left(\frac{x+1}{2}\right) + C_1\end{aligned}$$

and

$$-\frac{1}{2} \int \frac{2x+2}{x^2 + 2x + 5} dx = -\frac{1}{2} \ln|x^2 + 2x + 5| + C_2$$

by another u-sub.

***Finally*:**

$$\begin{aligned}\int \frac{-x+9}{x^2 + 2x + 5} dx &= -\frac{1}{2} \ln|x^2 + 2x + 5| \\ &\quad + 5 \arctan\left(\frac{x+1}{2}\right) + C\end{aligned}$$

Example #7 (Improper rational function)

Find

$$\int \frac{2x^2 + 5x + 6}{x^2 + 3x + 2} dx.$$

Divide

$$2x^2 + 5x + 6$$

by

$$x^2 + 3x + 2.$$

Here

$$\begin{array}{r} 2 \\ x^2 + 3x + 2 \overline{) 2x^2 + 5x + 6} \\ \underline{\ominus 2x^2 + 6x + 4} \\ -x + 2 \end{array}$$

tells us

$$2(x^2 + 3x + 2) + (-x + 2) = 2x^2 + 5x + 6$$

$$\Rightarrow 2 + \frac{-x + 2}{x^2 + 3x + 2} = \frac{2x^2 + 5x + 6}{x^2 + 3x + 2}$$

$$\frac{-x + 2}{x^2 + 3x + 2} = \frac{-x + 2}{(x + 1)(x + 2)} = \frac{A}{x + 1} + \frac{B}{x + 2}$$

$$-x + 2 = A(x + 2) + B(x + 1)$$

$$\text{If } x = -1$$

$$3 = A.$$

$$\text{If } x = -2$$

$$4 = -B \Rightarrow B = -4.$$

Now

$$\int \frac{-x+2}{(x+1)(x+2)} dx = 3 \int \frac{1}{x+1} dx - 4 \int \frac{1}{x+2} dx$$
$$= 3 \ln|x+1| - 4 \ln|x+2| + C$$

so

$$\int \frac{2x^2 + 5x + 6}{x^2 + 3x + 2} dx = \int 2 + \frac{-x+2}{(x+1)(x+2)} dx$$
$$= 2x + 3 \ln|x+1| - 4 \ln|x+2| + C.$$

Example #8 (Repeated irreducible factor)

Find the partial fraction decomposition of

$$\frac{1}{(x+1)(x^2+2x+2)^2}.$$

Guess the form

$$\frac{1}{(x+1)(x^2+2x+2)^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2+2x+2} + \frac{Dx+E}{(x^2+2x+2)^2}$$

then

$$1 = A(x^2+2x+2)^2 + (Bx+C)(x^2+2x+2)(x+1) + (Dx+E)(x+1).$$

Let $x = -1$

$$1 = A(1-2+2) \Rightarrow A = 1$$

Expand to find the other terms

$$\begin{aligned} 1 = & A(x^4 + 4x^3 + 8x^2 + 8x + 4) \\ & + B(x^4 + 3x^3 + 4x^2 + 2x) \\ & + C(x^3 + 3x^2 + 4x + 2) \\ & + D(x^2 + x) \\ & + E(x + 1) \end{aligned}$$

4th order terms ($A = 1$)

$$0 = (A + B)x^4 \Rightarrow B = -1$$

3rd order terms

$$0 = (4A + 3B + C)x^3 \Rightarrow 1 + C = 0 \Rightarrow C = -1$$

2nd order terms

$$\begin{aligned} 0 &= (8A + 4B + 3C + D)x^2 \\ \Rightarrow 8 - 4 - 3 + D &= 0 \Rightarrow D = -1 \end{aligned}$$

constant terms ($A = 1, B = -1, C = -1, D = -1$)

$$1 = 4A + 2C + E \Rightarrow E = -2C - 3 = 2 - 3 = -1$$

The PFD is

$$\frac{1}{(x+1)(x^2+2x+2)^2} = \frac{1}{x+1} + \frac{-x-1}{x^2+2x+2} + \frac{-x-1}{(x^2+2x+2)^2}.$$

Example #9

Find

$$\int \frac{1}{x\sqrt{1+2x}} dx.$$

Let

$$u^2 = 2x + 1 \quad 2u \, du = 2 \, dx \quad u \, du = dx$$

so

$$\Rightarrow x = \frac{u^2 - 1}{2} \quad u = (1 + 2x)^{1/2}.$$

Then

$$\begin{aligned} \int \frac{1}{x(1+2x)^{1/2}} dx &= \int \frac{2}{u^2 - 1} \cdot \frac{u}{u} du \\ &= \int \frac{2}{(u-1)(u+1)} du \end{aligned}$$

We have

$$\begin{aligned} \frac{2}{(u-1)(u+1)} &= \frac{A}{u-1} + \frac{B}{u+1} \\ 2 &= A(u+1) + B(u-1) \end{aligned}$$

If $u = 1$

$$\Rightarrow A = 1.$$

If $u = -1$

$$\Rightarrow B = -1.$$

$$\begin{aligned}\int \frac{2}{(u-1)(u+1)} du &= \int \frac{1}{u-1} du - \int \frac{1}{u+1} du \\ &= \ln|u-1| - \ln|u+1| + C \\ &= \ln \left| \frac{u-1}{u+1} \right| + C\end{aligned}$$

$$\Rightarrow \int \frac{1}{x(1+2x)^{1/2}} dx = \ln \left| \frac{(1+2x)^{1/2} - 1}{(1+2x)^{1/2} + 1} \right| + C$$

Example #10

Find

$$\int \frac{e^{3x} + e^{2x} + e^x}{(e^{2x} + 1)^2} dx.$$

Let

$$u = e^x \quad du = e^x dx.$$

so

$$(e^{3x} + e^{2x} + e^x)dx = (u^3 + u^2 + u)dx = (u^2 + u + 1)du.$$

We get

$$\int \frac{e^{3x} + e^{2x} + e^x}{(e^{2x} + 1)^2} dx = \int \frac{u^2 + u + 1}{(u^2 + 1)^2} du$$

and the PFD has the form

$$\frac{u^2 + u + 1}{(u^2 + 1)^2} = \frac{Au + B}{u^2 + 1} + \frac{Cu + D}{(u^2 + 1)^2}$$

$$u^2 + u + 1 = (Au + B)(u^2 + 1) + Cu + D$$

No 3rd order terms $\Rightarrow A = 0$, and we have a simpler system:

$$u^2 + u + 1 = B(u^2 + 1) + Cu + D$$

2nd order terms: $\Rightarrow B = 1$

1st order terms: $\Rightarrow C = 1$

constant terms:

$$B + D = 1 \Rightarrow D = 0$$

Then

$$\begin{aligned}\int \frac{u^2 + u + 1}{(u^2 + 1)^2} du &= \int \frac{1}{u^2 + 1} du + \int \frac{u}{(u^2 + 1)^2} du \\ &= \arctan(u) - \frac{1}{2} (u^2 + 1)^{-1} + C\end{aligned}$$

$$\Rightarrow \int \frac{e^{3x} + e^{2x} + e^x}{(e^{2x} + 1)^2} dx = \arctan(e^x) - \frac{1}{2} (e^{2x} + 1)^{-1} + C$$